

Question:

Carla is downloading a 200 GB file. Normally she can download 2 GB/minute, but 40% of the way through the download, Windows forces a restart to install updates, which takes 20 minutes. Then Carla has to restart the download from the beginning. How long does it take to download the file?

Reasoning + Answer:

<thinking process >

Step 1 : Understand the total size of the file and the normal download speed .

- The file is 200 GB .
- Normal download speed is 2 GB / minute .

Step 2 : Determine how much data Carla downloads before the restart .

- 40 % of the way through means she downloads 40 % of 200 GB = $0.40 \times 200 = 80$ GB .

Step 3 : Calculate how long it takes to download 80 GB at 2 GB / minute .

- Time = Amount / Speed = 80 GB / 2 GB / minute = 40 minutes .

Step 4 : The restart takes 20 minutes . During this time , the download is paused .

Step 5 : After the restart , she starts downloading again from the beginning , so she has to download the full 200 GB again .

Step 6 : Time to download the full 200 GB at 2 GB / minute .

- $200 \text{ GB} \div 2 \text{ GB / minute} = 100$ minutes .

Step 7 : Total time = time before restart + restart time + time to red ownload from start .

- 40 minutes (first part) + 20 minutes (restart) + 100 minutes (full download after restart) = 160 minutes .

Final answer : Total download time is 160 minutes .

</thinking process >

< answer > 160 </answer >

